

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Faculty: Dr. V. Parthipan

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I **M. Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

SHORT ANSWER TEST

Question 1

A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = 150 MB = 150,000,000 bytes
- Segment size = 1500 bytes
- Frame overhead (header + trailer) = 40 bytes/frame
- Bandwidth = 50 Mbps

Calculation:

Number of segments = File size / Segment size = $150,000,000 / 1500 = 100,000$ segments

Number of frames = 100,000 (one frame per segment) = 100,000 frames

Total DLL overhead = 100,000 frames $\times$ 40 bytes = 4,000,000 bytes = 4 MB

Total data transmitted = 150,000,000 + 4,000,000 = 154,000,000 bytes

Total data in bits = 154,000,000 $\times$ 8 = 1,232,000,000 bits

Transmission time = Total bits / Bandwidth = $1,232,000,000 / 50,000,000 = 24.64$ seconds

Result:

- Total segments = 100,000
- Total frames = 100,000
- Total DLL overhead = 4,000,000 bytes (4 MB)
- Total transmission time $\approx$ 24.64 seconds

Explanation:

The Transport Layer ensures reliable communication by breaking the file into manageable segments, numbering them sequentially, and using acknowledgements, retransmission of lost segments, and flow/congestion control (as in TCP) so that data reaches the destination in order and without loss. The Data Link Layer ensures reliability at the link level by adding framing information, performing error detection (using CRC/checksum in the trailer) and, where required, error correction or retransmission (ARQ), and by controlling access to the shared medium so that frames are delivered correctly between adjacent nodes.

Question 2

A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Window size = 6 frames
- Frame data size = 1024 bytes
- Total frames to send = 48
- One frame lost per window

Calculation:

Number of windows required = Total frames / Window size = $48 / 6 = 8$ windows

Frames lost = 1 per window $\times$ 8 windows = 8 lost frames

Each lost frame requires exactly one retransmission (assuming no further loss)

Result:

- Number of transmission windows = 8
- Total retransmissions = 8

Explanation:

Flow control (using the sliding window mechanism) allows the sender to transmit multiple frames before waiting for individual acknowledgements, which keeps the communication

channel busy and improves throughput. It prevents a fast sender from overwhelming a slower receiver's buffer, reduces idle time between transmissions, and, combined with selective/Go-Back-N retransmission, ensures that only the lost frames (rather than the entire window) need to be resent, improving overall efficiency.

Question 3

A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- Original packet size = 12,000 bytes
- MTU = 1500 bytes (per fragment, including header)
- IP header per fragment = 20 bytes

Calculation:

Usable data per fragment = $MTU - \text{Header} = 1500 - 20 = 1480$ bytes

Number of fragments = $\text{ceil}(12,000 / 1480) = \text{ceil}(8.11) = 9$ fragments

(8 fragments carry 1480 bytes each = 11,840 bytes; the 9th fragment carries the remaining 160 bytes)

Total header overhead = $9 \text{ fragments} \times 20 \text{ bytes} = 180$ bytes

Result:

- Total fragments created = 9
- Total header overhead = 180 bytes

Explanation:

The Network Layer ensures successful delivery of fragmented packets by assigning each fragment the same Identification number (so the destination knows which fragments belong together), a Fragment Offset (to indicate the position of each fragment's data in the original packet), and a More Fragments flag (cleared only on the last fragment). The destination host buffers incoming fragments and reassembles the original packet using this information before passing it up to the Transport Layer; routing and addressing at this layer ensure each fragment independently reaches the correct destination.

Question 4

A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Window size = 6 frames
- Frame data size = 1024 bytes
- Total frames to send = 48
- One frame lost per window

Calculation:

Number of windows required = $48 / 6 = 8$ windows

Frames lost = 1 per window $\times$ 8 windows = 8 lost frames

Each lost frame requires exactly one retransmission

Result:

- Number of transmission windows = 8
- Total retransmissions = 8

Explanation:

This question repeats Question 2 in the assessment paper. As explained above, flow control via the sliding window mechanism lets the sender transmit several frames before requiring acknowledgement, improving link utilisation, preventing receiver buffer overflow, and, along with the retransmission of only the lost frames, keeping the communication efficient even when isolated losses occur.

Question 5

A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created

before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- File size = 600 MB
- Checkpoint interval = every 60 MB
- Failure occurs after 390 MB transferred

Calculation:

Checkpoints occur at: 60, 120, 180, 240, 300, 360 MB (the next one, at 420 MB, is not reached)

Number of checkpoints created before failure = $\text{floor}(390 / 60) = 6$

Data since the last checkpoint (360 MB) at the time of failure = $390 - 360 = 30$ MB

Result:

- Checkpoints created before failure = 6
- Data to be retransmitted after recovery = 30 MB

Explanation:

Synchronization checkpoints inserted by the Session Layer allow a session to resume from the last successful checkpoint instead of restarting the entire transfer after a failure. This dialog control/recovery function saves time and bandwidth, maintains session state (such as which side transmits next and where the transfer left off), and is especially valuable for large, long-running transfers like multimedia streaming, ensuring continuity and reliability of the overall session.

Question 6

Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = 900 MB
- Compression reduces size by 40%

- Encryption adds 5% overhead to the compressed size

Calculation:

Compressed size = $900 \times (1 - 0.40) = 900 \times 0.60 = 540$ MB

Encrypted size = $540 \times (1 + 0.05) = 540 \times 1.05 = 567$ MB

Total reduction compared to original = $900 - 567 = 333$ MB

Percentage reduction = $(333 / 900) \times 100 = 37\%$

Result:

- Compressed file size = 540 MB
- Encrypted file size = 567 MB
- Overall reduction compared to the original file = 37%

Explanation:

The Presentation Layer improves transmission efficiency by compressing data, which reduces the volume of data sent over the network, lowering bandwidth consumption and transmission time. It improves security by encrypting the data before transmission, protecting confidentiality and integrity from eavesdropping or tampering while in transit. It also handles translation/formatting (e.g., character encoding, data representation) so that data sent by one system is correctly interpreted by another, regardless of underlying platform differences.

Question 7

An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Total data = 3 GB = 3000 MB
- Request size = 3 MB
- Effective throughput = 120 Mbps

Calculation:

Total requests = $3000 \text{ MB} / 3 \text{ MB} = 1000$ requests

Total data in bits = $3 \text{ GB} \times 1000 \text{ (MB/GB)} \times 8 \text{ (bits/byte)} = 3000 \times 8 = 24,000 \text{ Mb}$

Transmission time = Total bits / Throughput = 24,000 Mb / 120 Mbps = 200 seconds

Result:

- Total requests generated = 1000
- Total transmission time $\approx$ 200 seconds

Explanation:

The Application Layer provides reliable services to users through protocols such as FTP that break large transfers into manageable requests/responses, use acknowledgements to confirm each request was received, and support retry of failed requests. It presents a simple, user-facing interface (commands like GET/PUT) while relying on the underlying Transport Layer (e.g., TCP) for guaranteed, in-order, error-checked delivery, giving the end user a dependable file-transfer experience.

Question 8

A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Processing delays: Application = 4 ms, Transport = 3 ms, Network = 5 ms, Data Link = 2 ms, Physical = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

Calculation:

Total processing delay = 4 + 3 + 5 + 2 + 1 = 15 ms

Total end-to-end delay = Processing delay + Propagation delay + Transmission delay

Total end-to-end delay = 15 + 18 + 12 = 45 ms

Result:

- Total end-to-end delay = 45 ms

Explanation:

Each OSI layer adds its own processing time as the packet moves down the sender's stack and up the receiver's stack: the Application Layer prepares user data and requests services; the Transport Layer segments data and manages reliability/flow control; the Network Layer performs routing and logical addressing; the Data Link Layer handles framing, addressing and error detection; and the Physical Layer converts frames into signals for transmission. In addition to these per-layer processing delays, the transmission delay (time to push all bits onto the link) and propagation delay (time for a signal to travel the physical distance) both contribute to the total end-to-end delay experienced by the packet.

Question 9

A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead per frame = 80 bytes

Calculation:

Useful data per frame = Frame size – Overhead = 1600 – 80 = 1520 bytes

Number of frames required = $\text{ceil}(80,000,000 / 1520) = 52,632$ frames

Total overhead transmitted = $52,632 \times 80 = 4,210,560$ bytes

Transmission efficiency = Useful data per frame / Frame size $\times 100 = 1520 / 1600 \times 100 = 95\%$

Result:

- Total frames required $\approx 52,632$
- Total overhead transmitted $\approx 4,210,560$ bytes
- Transmission efficiency = 95%

Explanation:

Protocol overhead (headers/trailers added by each OSI layer) consumes part of every frame's capacity without carrying useful data, so it directly reduces transmission efficiency and effective throughput. As overhead increases relative to frame size, more frames or more bandwidth are needed to deliver the same amount of useful data, increasing transmission time and network load — this is why protocols try to keep frame sizes reasonably large relative to header overhead to maximise efficiency.

Question 10

A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original file size = 240 MB
- Compression reduces size by 30%
- Transport Layer segment size = 1400 bytes
- Data Link Layer overhead = 60 bytes/frame

Calculation:

Compressed size = $240 \times (1 - 0.30) = 240 \times 0.70 = 168 \text{ MB} = 168,000,000 \text{ bytes}$

Number of segments = $168,000,000 / 1400 = 120,000 \text{ segments}$

Total Data Link overhead = $120,000 \times 60 = 7,200,000 \text{ bytes}$

Total data transmitted at Physical Layer = $168,000,000 + 7,200,000 = 175,200,000 \text{ bytes} \approx 175.2 \text{ MB}$

Result:

- Compressed file size = 168 MB
- Total segments transmitted = 120,000

- Total protocol overhead = 7,200,000 bytes (≈ 6.87 MB)
- Final data transmitted at the Physical Layer ≈ 175.2 MB

Explanation:

Multiple OSI layers cooperate to make this transfer both efficient and trustworthy: the Presentation Layer compresses (and can encrypt) the sensitive medical data to save bandwidth and protect patient confidentiality; the Transport Layer segments the compressed data and provides reliable, ordered, end-to-end delivery with error recovery; the Network Layer routes segments across the IoT healthcare network to the correct destination; the Data Link Layer frames each segment and adds overhead for addressing and error detection over each physical hop; and the Physical Layer transmits the final bit stream over the medium. Together these layers ensure the patient record is delivered completely, in order, securely, and efficiently despite the constraints of the healthcare network.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation